

Juan Esteban Villegas Delgado, Ph.D.

Senior Quantum Hardware & Microfabrication Engineer

Ph.D. Electrical Engineering | M.Sc. Microsystems Engineering | B.Sc. Mechatronics Engineering
Quantum Hardware Lead Designer & Postdoctoral Researcher, Technology Innovation Institute, UAE

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Abu Dhabi

United Arab Emirates

EXECUTIVE PROFILE

Senior Quantum Hardware Engineer and Fabrication Reviewer with 12+ years of experience spanning industrial systems engineering, multidisciplinary team leadership, and full-stack development of superconducting quantum processing units (QPUs) and integrated photonic chips. In my current role, I lead superconducting microwave circuit design and serve as primary fabrication reviewer—bridging electromagnetic/FEM layout with wafer-scale Design-for-Manufacturing (DFM), process window optimization, and device reliability. I drive thin-film and Josephson junction process qualification across Aluminum, Niobium, and Tantalum platforms, tracking critical temperature (T_c), sheet resistance uniformity, critical current density (J_c) mapping, dielectric loss-tangent suppression, and long-term resonator Q -factor and qubit coherence stability through full multi-qubit QPU bring-up and automated calibration.

Backed by a strong industrial foundation leading multidisciplinary engineering teams in commercial and naval vessel design, I excel at translating complex physics into manufacturable hardware under strict delivery schedules. In ISO Class-5 cleanroom and cryogenic laboratory operations, I author and audit Standard Operating Procedures (SOPs), train technicians and engineers in nanofabrication and metrology protocols, and integrate emerging 3D interconnect (airbridge) and hybrid manufacturing processes.

At the operational and management level, I oversee hardware project schedules, cleanroom equipment procurement, vendor preventative maintenance/service contracts, and external supplier/foundry deliverables. I actively support organizational scaling by conducting technical interviews for entry- and mid-level engineering candidates, mentoring staff, and aggregating project KPIs for formal performance reviews. Fluent in English, Spanish, and French.

TECHNICAL SKILLS

Quantum Hardware Design, EDA & AI-Driven Automation: Full-stack circuit quantum electrodynamics (cQED) and integrated photonic architecture; Hamiltonian modeling and energy-participation quantization (`pyEPR`, `scqubits`, `QuTiP`); programmatic physical layout generation, PDK development, and automated Design-for-Manufacturing (DFM) verification (`KLayout` Python API); AI/ML-assisted inverse design, automated parameter sweeps, and closed-loop electromagnetic/optical design-space optimization.

QPU Bring-Up, Cryogenic RF & Automated Calibration: End-to-end cryogenic RF/microwave packaging and multi-qubit QPU characterization below 20 mK; operation, troubleshooting, and preventative maintenance of dilution refrigerators and ultra-high vacuum (UHV) systems; development of automated data-acquisition, pulse-level control, and AI-integrated QPU calibration and benchmarking pipelines (`Qibo`/`Qibocal`, `Quantify`, `Qiskit`) for single- and two-qubit gate tuning, crosstalk mitigation, and automated tracking of qubit coherence (T_1, T_2^*) and resonator internal quality factors (Q_i).

Cleanroom Microfabrication & Process Control: Wafer-scale process develop-

ment, fabrication readiness review, and yield optimization for superconducting (Al, Nb, Ta) and silicon photonic (Si, Si₃N₄) platforms; Dolan/Manhattan shadow-evaporated Josephson tunnel junctions, 3D micro-airbridges, and heterogeneous 2D material integration; thin-film deposition (e-beam and thermal evaporation, magnetron sputtering, ALD, PECVD, LPCVD, electroplating), dry/wet etching (ICP-RIE, DRIE, FIB, wet chemical etching), and lithography (Electron-Beam, laser direct-write, and optical lithography); cleanroom SOP authoring and additive manufacturing.

Photonic, Superconducting & Material Metrology: Room-temperature and cryogenic wafer probing, four-point sheet-resistance and critical current density (J_c) mapping, critical temperature (T_c) and dielectric loss-tangent characterization; high-speed opto-electronic testing, Optical Vector/Network Analysis, insertion-loss and polarization metrology, and DC I - V curve tracing; Scanning Electron Microscopy (SEM-EDS), Atomic Force Microscopy (AFM), X-Ray Diffraction (XRD), Raman spectroscopy, spectroscopic ellipsometry, optical interferometry/profilometry, ICP-MS, and Ion Chromatography.

Programming & Software Engineering: Python (scientific computing, hardware control, ML/automation), C++, C#, MATLAB, LabVIEW, Git, Docker, Linux.

Simulation & Quantum Frameworks: Ansys (HFSS, Q3D), COMSOL Multiphysics, Lumerical (FDTD, MODE), MEEP, QuTiP, scqubits, pyEPR, Qiskit, Qibo (qibolab/qibocal), Quantify.

CAD & Physical Layout: KLayout (scripted EDA/PCell automation), SolidWorks, Autodesk Inventor, AutoCAD, Rhinoceros.

WORK EXPERIENCE

Postdoctoral Researcher — Superconducting Circuit Lead Designer & Fabrication Reviewer

Technology Innovation Institute

September 2023 - to date

Abu Dhabi, UAE

Lead the full-stack microwave circuit design, fabrication review, and cryogenic bring-up of multi-qubit superconducting Quantum Processing Units (QPUs), alongside operational management of an ISO Class-5 cleanroom and dilution refrigerator facility (1.65 mW combined cooling power at 20 mK).

- **Circuit Design & Fabrication Readiness Review:** Lead the electromagnetic/FEM design and automated physical layout of multi-qubit superconducting processors; serve as primary technical fabrication reviewer enforcing Design-for-Manufacturing (DFM) rules, wafer-level yield optimization, and process window control. Design and develop EDA integration tools for superconductive circuits. Develop and coordinate trainings for QPU design, fabrication and calibration.
- **Superconducting Process Control & Reliability Metrics:** Drive thin-film (Al, Nb, Ta) and Josephson junction process qualification, tracking film conductivity and critical temperature (T_c), sheet resistance uniformity, critical current density (J_c) wafer mapping, dielectric loss-tangent reduction, and long-term resonator Q -factor and qubit coherence stability.
- **QPU Bring-Up & Automated Calibration:** Lead cryogenic RF packaging, cooldown characterization, and full-stack QPU bring-up, developing automated single- and two-qubit calibration and benchmarking pipelines.
- **Cleanroom Operations, SOPs & Training:** Author, review, and enforce cleanroom Standard Operating Procedures (SOPs) for lithography, etching, thin-film deposition, and 3D airbridge integration; train and supervise technicians and junior engineers on tool operation, safety, and metrology protocols.
- **Procurement, Suppliers & Service Contracts:** Manage cleanroom and cryogenic capital/consumable procurement, qualify external foundries and specialty

material suppliers, and oversee vendor preventative maintenance and service contracts for critical fabrication and cryogenic infrastructure.

- **Project Governance, Hiring & Performance Reviews:** Manage multi-project hardware delivery schedules, conduct technical screenings and interviews for entry- and mid-level engineering candidates, mentor staff, and aggregate project KPIs and technical milestones for formal performance reviews.

Graduate Research Intern

July 2022 - October 2022

Technology Innovation Institute

Abu Dhabi, UAE

Designed, taped out, and characterized integrated Silicon Nitride (Si_3N_4) photonic chips for quantum single-photon sources and commercial-grade Quantum Random Number Generators (QRNGs), coordinating foundry DFM compliance and optical test-bench automation.

Technology and Data Leader

January 2018 - June 2020

Airborne

Abu Dhabi, UAE

Directed the product engineering and hardware/software integration division developing commercial indoor navigation systems for visually impaired users. Managed product roadmaps, cross-functional engineering schedules, client and stakeholder requirements, and end-to-end prototype-to-product validation.

Research Engineer

August 2018 - April 2019

Khalifa University

Abu Dhabi, UAE

Led analytical metrology and high-resolution physical/chemical characterization workflows using SEM-EDS, Ion Chromatography (IC), and ICP-MS. Established laboratory quality-control SOPs, calibrated analytical instrumentation, and authored technical compliance reports.

Development Engineer

June 2015 - July 2016

Bureau Veritas

Antwerp, BE

Authored statutory engineering rules, regulatory SOPs, and technical standards for commercial inland navigation vessels, specializing in shipboard electrical distribution, automation, and electronic control systems. Interfaced directly with shipyards, equipment manufacturers, and regulatory bodies, and developed and delivered technical certification courses for industry professionals.

Marine Designer and Researcher —

COTECMAR

Engineering Team Lead

February 2012 - June 2015

Cartagena, CO

Led multidisciplinary engineering teams across the Naval Architecture and Electrical/Electronics divisions for the industrial design, manufacturing, and commissioning of high-speed commercial and patrol vessels.

- **Team & Schedule Leadership:** Managed engineering design teams, master project schedules, and cross-departmental conflict resolution between design, shipyard production, and quality assurance units.
- **Supplier, Procurement & Client Management:** Defined technical specifications for shipboard electronics and power systems, evaluated vendor bids, managed supplier deliverables, and led client design reviews and sea-trial acceptance testing.
- **R&D & Prototyping:** Led applied R&D in ruggedized naval electronics and telecommunications, taking tactical marine datalink protocols from architecture through hardware prototyping and field deployment.

**CURRENT
PROJECTS**

New Materials for Superconductive Quantum Electrodynamics: 2023-Current

- Process development, thin-film growth qualification, and cryogenic RF characterization of high-impedance (high kinetic inductance L_k) superconducting materials for next-generation QPUs, targeting sheet-resistance uniformity, critical temperature (T_c)

control, and low dielectric loss.

Principal Investigator: Dr. Frederico Brito, TII, UAE.

Design and Test of Superconductive Quantum Processing Units: *2023-Current*

- Full-stack circuit quantum electrodynamics (cQED) architecture, FEM simulation, and fabrication review of multi-qubit QPUs—including high- Q coplanar waveguide resonators, tunable and fixed couplers, 3D airbridges, and Al/AIO_x/Al Josephson tunnel junctions (J_c targeting and yield optimization).

Principal Investigator: Dr. Frederico Brito, TII, UAE.

SELECTED COMPLETED PROJECTS

8-Qubit Fixed-Frequency All-Microwave Superconducting QPU: *2025-2026*

- Led the end-to-end Hamiltonian parameter calculation, electromagnetic simulation, and physical layout of an 8-qubit fixed-frequency all-microwave quantum processor. Coordinated cleanroom manufacturing, DFM reviews, and wafer qualification with the fabrication team; led cryogenic bring-up with the measurement lab team and executed calibration experiments to produce high-fidelity single- and two-qubit microwave gates.

Direct Manager: Dr. Frederico Brito, TII, UAE.

Low-Loss Superconducting Platform Based on Tantalum (Ta) Encapsulation: *2025-2026*

- Designed experimental test vehicles and microwave resonator metrology structures to evaluate Tantalum-encapsulated superconducting thin films aimed at suppressing surface oxide formation and two-level system (TLS) dielectric losses. Led experimental design, oversaw thin-film process characterization (T_c , sheet resistance, and internal quality factor Q_i stability), and reviewed scientific data analysis.

Direct Manager: Dr. Frederico Brito, TII, UAE.

Quantum Computing Cryogenic & Cleanroom Infrastructure: *2023-2024*

- Commissioned, qualified, and established operational SOPs for the dilution refrigerator measurement facility and cleanroom test infrastructure supporting the first domestically developed quantum processors in the UAE.

Principal Investigators: Prof. Andrew Lutken and Dr. Frederico Brito, TII, UAE.

High-Speed Mode and Polarization Encryption for Optical Communication Networks: *2021-2023*

- Designed and fabricated integrated silicon photonic devices and high-speed optical testbeds for mode- and polarization-division multiplexed encryption and Optical Physical Unclonable Function (OPUF) key distribution.

Principal Investigator: Dr. Mahmoud Rasras, NYUAD, UAE.

Heterogeneously Integrated Layered Materials for Silicon Photonics: *2021-2024*

- Designed and fabricated planar silicon photonic circuits engineered for back-end-of-line (BEOL) heterogeneous integration with 2D layered materials for high-speed photodetectors and modulators.

Principal Investigator: Dr. Mahmoud Rasras, NYUAD, UAE.

Plasmonic-Enhanced Physically Unclonable Functions: *2020-2021* - Developed nanofabrication processes to integrate plasmonic resonant structures onto silicon photonic waveguides, generating tamper-evident stochastic scattering responses for hardware cryptography.

Principal Investigator: Prof. Mahmoud Rasras, NYUAD, UAE.

PM2.5 Airborne Pollutant Characterization Over the UAE: *2018-2019* - Led multi-site sampling, SEM-EDS/ICP-MS metrology, and source apportionment of PM2.5 particulates across the UAE. Authored policy recommendations for government environmental agencies and trained technical personnel on standardized measurement SOPs.

Direct Manager: Prof. Marouane Temimi, Khalifa University, UAE.

On-Board Load Computers Technical Specification: *2015-2016* - Authored the

statutory classification rules and certification test standards for shipboard load/stability computers and marine electronic systems across European inland waterways.

Direct Manager: Mr. Jean Michel Chatelier, Bureau Veritas, Belgium.

Naval Datalink System: 2014-2015 - Led system architecture, supplier specification, and hardware/firmware prototyping for a tactical naval datalink platform (OSI layers 2–6) deployed by the Colombian Navy.

Direct Manager: (R) Rear-Admiral LH (OF-6) Gustavo Perez, COTECMAR - ARC, Colombia.

Riverine Patrol Craft (LPR-MkII): 2012-2014 - Led multidisciplinary engineering design covering updated hull hydrodynamics, structural layout, electrical/electronic system integration, and shipyard production support for a rapid riverine patrol craft series.

Direct Manager: Commander (OF-4) Ricardo Mejía, COTECMAR - ARC, Colombia.

EDUCATION

Escuela de Ingeniería de Antioquia (EIA) University, Medellín, ANT, Colombia
Bachelor of Sciences, Mechatronics Engineering, December 2011

Thesis: “Model of solar powered water pumping systems for small distribution networks on rural arid regions in Colombia”, work carried with the Sustainability Research Group at EIA University

Khalifa University (Masdar Institute), Abu Dhabi, AD, United Arab Emirates
Master of Science, Microsystems Engineering, May 2018

Thesis: “Studies of materials for planar integrated photonics: PDMS and amorphous Silicon”, work carried at the Masdar Institute, UAE, in collaboration with MIT.

New York University, New York City, USA & Abu Dhabi, UAE
Philosophy Doctor, Electrical Engineering, October 2023

Thesis: “Integrated Silicon Photonics for Physical Layer Security Applications”, work carried at the Center for Cyber Security at NYU Abu Dhabi.

TRAININGS

Superconductive Qubit Design, CMC Microsystems, Vancouver, Canada (December 2024).

Silicon Photonics Design, CMC Microsystems, Vancouver, Canada (August 2022).
Silicon Photonics Design, Fabrication and Data Analysis, UBCx (November 2019).

Train the Trainer: Management & Development of Technical Trainings, Opatija, Croatia (March 2016).

Inland Navigation Statutory Engineering Regulations, Bureau Veritas, Antwerp, Belgium (November 2015).

Electromagnetic Compatibility (EMC) in Engineering Systems, Cartagena, Colombia (June 2014).

LEADERSHIP & AFFILIATIONS

Technical Training & Mentorship: Extensive experience training cleanroom technicians and junior engineers on nanofabrication SOPs and cryogenic metrology (TII, UAE), delivering statutory engineering certification courses to industry professionals (Bureau Veritas, Belgium), and serving as a graduate/undergraduate instructor in Photonics and Computer Programming for Engineers (NYU Abu Dhabi & Khalifa University).

Engineering Governance & Operations: Founding member and Chair (Spring 2023) of the Early Career Researchers Forum (EERF) and Graduate Council representative (2019–2022) at NYU Abu Dhabi. Former National Coordinator of Operations & Volunteering in Brussels, Belgium (2015–2016) and Construction & Logistics Manager in Cartagena, Colombia (2014–2015) for TECHO.

Professional Memberships: Member of the IEEE Photonics Society, Optica, and

the Colombian Optical Society.

PROFESSIONAL REFERENCES **Prof. Dr. Frederico Brito**, Director of Quantum Hardware, Quantum Research Center, TII, Abu Dhabi, UAE
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